MEDOC changes log

Medoc V2.4 update

Bug correction

Small bug in the computation of non site specific energy corrected :

A kept state was double counted, being also present in the discreded energy.

State pruning is now a specific option, pruning after or during no longer needs to be specified

Comments were updated to reflect and mention the publication

Comments now refer to the equations in the publication

Variable names were changed to match the indices in the publication

Figure file name details is now an option

If is turned on, the name of the figure files will contain some important execution option such as SS (site specific), nneigh, T, ect… Turn on using -ds (detailed_suffix)

Two chokeholds where resolved

These two were plaguing the program especially for lower precision systems ## The limitation on the free energy range is solved In the get_G_sum function, the values are first gotten rid of positive infinities, (since $-\ln\left(e^{-A}+e^{-\infty}\right)=A$), then we substract the first element of the list to contain the values within smaller values, thus avoiding overflow errors.

So $G_{A+B}=-RT\ln\left(e^{0}+e^{-\frac{G_B-G_A}{RT}}\right)+G_A$

The limitation on the partition function

This error was due to the limitation having to compute all the weights before the partition function. To avoid this we are omputing the invert of the probabilities instead.

$$p_{i,(pH)}=\frac{e^{-\frac{\Delta F_{i,(pH)}}{RT}}}{\sum_j^N e^{-\frac{\Delta F_{j,(pH)}}{RT}}}=\frac{1}{e^{+\frac{\Delta F_{i,(pH)}}{RT}}\sum_j^N e^{-\frac{\Delta F_{j,(pH)}}{RT}}}=\frac{1}{\sum_j^N e^{-\frac{\Delta\Delta F_{i-j,(pH)}}{RT}}}$$

Where

$$\Delta\Delta F_{i-j,(pH)} = \Delta\Delta F^0_{i-j} + \Delta\Delta n_{i-j} pH \times RT \ln 10$$

Which, by limiting ourselves to differences, severely limits the range of the value and thus circumvents the very large partition function problems

Please do not hesitate to contact me if you have question, found a bug, or want to use MEDOC for specific purpose but don't know how (martin.fossat@gmail.com;fossat@ie-freiburg.mpg.de)